1 Conditions for the Existence of Left and Right Inverses

Let $A \in M_{m\times n}$
To establish the conditions for the existence of left and right inverses, we introduce the following notation:

Definition 1.

$$\begin{aligned} \mathrm{Col}(A) = \{Ax \in \mathbb{R}^m : x \in \mathbb{R}^n\} \quad \mathrm{Col}(A) \subseteq \mathbb{R}^m \quad \textit{then} \quad \mathrm{rank}(A) \leq m \\ N(A) = \{x \in \mathbb{R}^n : Ax = 0 \in \mathbb{R}^m\} \quad N(A) \subseteq \mathbb{R}^n \quad \textit{then} \quad \textit{nulity}(A) \leq n \end{aligned}$$

Theorem 1. $n = \mathrm{rank}(A) + dim(N(A))$

which implies that

$$n = \mathrm{rank}(A) \leq m \iff \textit{nulity}(A) = 0$$

$$\mathrm{rank}(A) = m \iff m \leq n$$

Proposition 1. *A has left inverse* $\implies N(A) = \{0\}$

Proof. Suppose that $A_l^{-1} \in M_{n\times m}$ is a left inverse of A. Then

$$A_l^{-1}A = I_n.$$

Let $x \in N(A)$. Since $Ax = 0$, it follows that

$$\begin{aligned} x &= I_n x \\ &= A_l^{-1}Ax \\ &= A_l^{-1}(0) \\ &= 0 \end{aligned}$$

Hence,

$$N(A) = \{0\}.$$

Furthermore, by the rank-nullity theorem,

$$n = \mathrm{rank}(A) \leq m$$

□

Proposition 2. *If* $N(A) = \{\mathbf{0}\}$*, then A has left inverse*

Proof. Suppose that $A^\top Ax = \vec{0}$, then,

$$\begin{aligned} x^\top A^\top Ax &= (Ax)^\top Ax \\ &= \|x\|^2 \\ &= Ax \\ &= \vec{0} \end{aligned}$$

Therefore, $N(A^\top A) = N(A) = \{0\}$.

Since $A^\top A$ is an $n \times n$ square matrix, the rank-nullity theorem implies that $\text{rank}(A^\top A) = n$.

$$(A^\top A)^{-1} A^\top A = ((A^\top A)^{-1} A^\top)A = I_n$$

Therefore,

$$(A^\top A)^{-1} A^\top \text{ is a left inverse of A}$$

□

Proposition 3. *If* $\text{rank}(A) = m$*, then A has a right inverse.*

Proof. Let y be an arbitrary vector in $\mathbb{R}^m$.

Since $\text{rank}(A) = m$, there exists $x \in \mathbb{R}^m$ such that $Ax = y$, which means $\text{Col}(A) = \mathbb{R}^m$.

Since $\text{rank}(A) = \text{rank}(A^\top) = m$ and $m = \text{rank}(A^\top) + nulity A^\top$.

$$N(A^\top) = \{0\}$$

Suppose that $AA^\top = 0$. Then

$$x^\top AA^\top = \|A^\top x\|^2 = 0$$

$$N(AA^\top) \subseteq N(A^\top)$$

Since $N(A^\top) = \{0\}$, it follows that $N(AA^\top) = \{0\}$.

By using the rank-nulity theorem,

$$\text{rank}(AA^\top) = m$$

Because $AA^\top$ is a square matrix, it is therefore invertible.

$$AA^\top (AA^\top)^{-1} = A(A^\top (AA^\top)^{-1}) = I_m$$

Hence, $A^\top (AA^\top)^{-1}$ is right inverse of A □